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# Our Universe in Brief

Our universe is vast. Achingly beautiful. Remarkably simple in some ways, intricately complex in others. From our universe's great richness, we'll need only a few basic facts that I'll now lay bare.

## The Big Bang

Our universe was born in a gigantic explosion 13.7 billion years ago. The explosion was given the irreverent name "the big bang" by my friend Fred Hoyle, a cosmologist who at that time (the 1940s) thought it an outrageous, fictional idea.

Fred was proved wrong. We've since seen radiation from the explosion, even in just the last week (as I write this) tentative evidence for radiation emitted in the first trillionth of a trillionth of a trillionth of a second after the explosion began![2]

We don't know what triggered the big bang, nor what, if anything, existed before it. But somehow the universe emerged as a vast sea of